Text Classification

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Learning paradigms for NLP tasks

- An NLP task belongs to one of the following three learning paradigms
  - Sequence-Labelling ⇒ $n >= 1$ and $\boldsymbol{n} == \boldsymbol{m}$
  - Classification/Regression ⇒ $n >= 1$ and $\boldsymbol{m} == \mathbf{1}$
  - Sequence Transformation (Generation) ⇒ $n >= 1$ and $\boldsymbol{m} >= \mathbf{1}$

$$f(x_1, x_2, \ldots, x_n) \quad \Rightarrow \quad y_1, y_2, \ldots, y_m$$

Sequence-labelling Tasks	Classification Tasks	Generation Tasks
• PoS tag	• Sentiment Classification	• Machine Translation
• NER	• Hate speech Classification	• Summarization
• Answer Extraction	• Fake News Detection	• Answer Generation

Text Classification

$$f(x_1, x_2, \ldots, x_n) \Rightarrow y \quad \text{, where } y \in \{y_1, y_2, \ldots, y_k\}$$

- Multiclass classification task:
  - $|y| = 1$

f(The movie was good.) ⇒ positive

- Multilabel classification task:
  - $|y| >= 1$

f(Good movie with terrible songs.) ⇒ {positive, negative}

f(.) can be any machine learning classifier, e.g., NB, DT, SVM, CNN, RNN, etc.

Sentiment Analysis

Sentiment Analysis

- It aims to identify the orientation of opinion (i.e., *positive*, *negative*, *neutral* or *conflict*) from natural language text using computational methods.

Why SA in important?

- **Human Psychology:** What others think and feel about an entity influences our understanding as well.

- Advantages
    - Businesses refine their products and services based on users' feedback.
    - Individual: Assist in making a decision, whether to acquire a service or purchase a product or not.
    - Economic forecasting / Stock price predictor
    - Understanding societal views.

Sentiment Analysis: Formal definition

Example:

Good product. Value for money phone. Everything is average. Display is good. Battery life is also good. I bought it for a employee of mine. Overall a good phone. 4/64 combo is hard to find in this price range. Exceeded my expectations.

- Four fundamental questions:
    - What’s the sentiment?
    - Who holds the sentiment?
    - About what?
    - When?

Sentiment Analysis: Formal definition

- Expressed as quintuple

  <*document*, *sentiment*, *holder*, *entity*, *time*>

- *Document*
  - Input text (a *phrase*, a *sentence*, a *paragraph*, a *document*)
- *Sentiment orientation*
  - Categorical (*positive*, *negative*, *neutral*, *conflict*) or
  - Ordinal (*weak positive*, *mild positive*, *strong positive*)
- *Sentiment Holder*
  - Usually, author of the *document*.
- *Target Entity*
  - *Product*, *service*, *topic*, *person*, *organization*, *issue*, or *event*
- *Sentiment Time*

Sentiment Analysis: Subproblems

Subproblems	Text	Remarks
Subjectivity	*This movie is **awesome**.*	Positive
	*This movie is **pathetic**.*	Negative
	*This movie is **3-hours long**.*	Neutral
Thwarting	***Impressive** story, **good** acting, however, it **didn't meet** my expectation.*	Small portion at the end dictates its sentiment.
Sarcasm	*This movie is **awesome to put you to sleep**.*	Criticism in a humorous way.
Humble Bragging	*My life is **miserable**, I have to **sign 300 autographs per day**.*	Draw attention to something of which someone is proud.
Discourse-based SA	*This movie is a classic, **although**, I don't like 'sci-fi'.*	Sentiment is altered due to connectives.
Sense-based SA	*Shane Warne is a **deadly** spinner. (Positive)*	Different senses lead to different sentiments.
	*The campus has **deadly** snakes. (Negative)*	
Sentiment Intensity	*Movie was **ok**.*	Weak positive sentiment.
	*Movie was **good**.*	Mild positive sentiment.
	*Movie was **awesome**.*	Strong positive sentiment.

Related Concepts

- Opinion, evaluation, appraisal, attitude, affect, emotion, mood, etc.
  - Related terms but technically they are different

Muzero et al., 2014. ***Are They Different? Affect, Feeling, Emotion, Sentiment, and Opinion Detection in Text*****.**
https://ieeexplore.ieee.org/document/6797872

Sentiment vs Opinion

- Sentiment or opinion?
    - *It has a great camera.*
    - *I love blue-colored phone.*
    - *I am concerned about the current state of the economy.*
    - *I think the economy is not doing well.*

- Sentiment expresses a more generic/societal view
- Opinion are personal interpretations.

Have you seen this movie?

Inside out (2015)

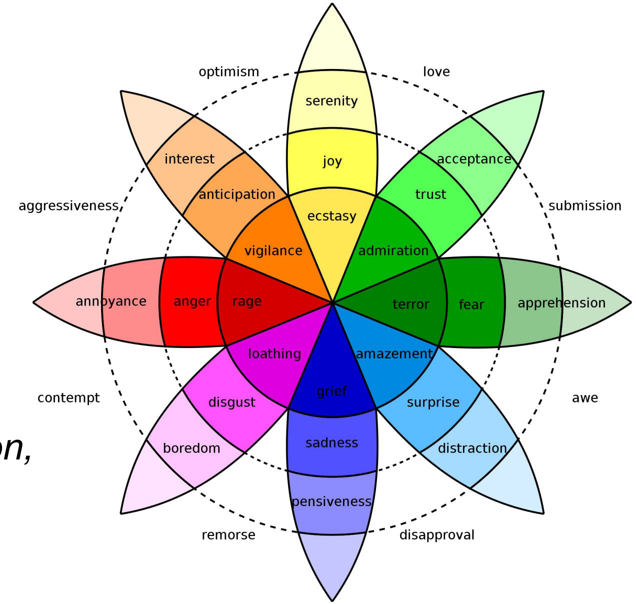
Emotion Analysis

- Study of the feeling or psychological state of a human mind.

    Just died from laughter after seeing that. → *joy*
    Still salty about that fire alarm at 2am this morning. → *fear*

- Ekman's emotion model [Ekman, 1982]
    - Six basic human emotions
        - *anger*, *joy*, *disgust*, *fear*, *surprise*, and *sadness*

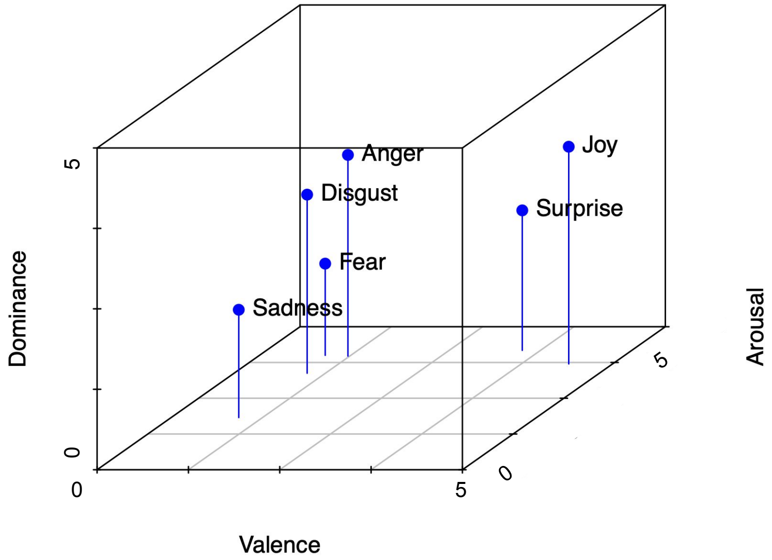
- Plutchik's 'wheel-of-emotions' [Plutchik, 1980]
    - Eight primary emotions
        - *anger*, *joy*, *disgust*, *fear*, *surprise*, *sadness, anticipation*, and *trust*



Emotion Analysis

- Russell and Mehrabian's VAD model
  - **Valence**: Expresses pleasant/unpleasant emotional scenario
  - **Arousal**: Measures the degree or intensity of the emotion
  - **Dominance**: Signifies control over the emotional state

	V	A	D
I am thrilled with the price.	4.4	4.4	4.0
I hate it, despise it, abhor it!	1.0	4.4	2.2
I shivered as I walked past the pale man's blank eyes, wondering what they were staring at.	1.2	3.0	1.5



Sentiment Classification

Sentiment Classification

- Given a $document$, identify the expressed $sentiment$
  - Assumptions
    - $holder$, $entity$, and $time$ are either known or simply ignore them.
- Similar to text classification.
  - Similarity
    - $f(X \in \mathbb{R}^d) \rightarrow C$
  - Dissimilarity
    - Important/cue words (features) for text classification are topic-related words
      - Document classification for sports → Cricket, Sachin, Bat, Pitch, etc.
    - Important/cue words (features) for sentiment classification are sentiment-bearing words (e.g., adjectives)
      - *I am concerned about the current state of the economy.*
- How does a model know that *concerned* is associated with negative sentiment?
  - Sentiment Lexicons

Sentiment Lexicons

- A dictionary of sentiment-bearing words with associated polarity labels
  - Good, admire, awesome, great, fine, etc. → *positive*
  - Bad, terrible, poor, overpriced, miserable, etc. → *negative*

- Harvard General Inquirer [Philip Stone, 1966] → ~4000 words
- Bing Liu [Hu and Liu, 2004] → ~6800 words
- MPQA [Wilson et al., 2005] → ~8000 words
- NRC Emotion Lexicon [Mohammad and Turney, 2010] → ~14000 words
- SentiWordNet [Baccianella, 2010] → WordNet synsets (~1.75 lakhs)
- NRC Hashtag Sentiment Lexicon [Mohammad et al., 2013]. → 54129 unigrams & 316531 bigrams.
- Sentiment140 [Mohammad et al., 2013] → 62468 unigrams & 677698 bigrams
- ...

Manually curated vs *Automatically processed*

Unsupervised classification [Turney, 2002]

- Three steps
  - Identify adjectives or adverbs phrases
  - Calculates the *semantic orientation* (SO) *scores* of the phrases
  - Aggregate SO to classify the reviews into *recommended* or *non-recommended* classes.

	First Word	Second Word	Third Word (Not Extracted)
1.	JJ	NN or NNS	anything
2.	RB, RBR, or RBS	JJ	not NN nor NNS
3.	JJ	JJ	not NN nor NNS
4.	NN or NNS	JJ	not NN nor NNS
5.	RB, RBR, or RBS	VB, VBD, VBN, or VBG	anything

$$\mathrm{PMI}(word_1, word_2) = \log_2\left(\frac{p(word_1 \,\&\, word_2)}{p(word_1)\, p(word_2)}\right)$$

$$\mathrm{SO}(phrase) = \mathrm{PMI}(phrase, \text{"excellent"}) - \mathrm{PMI}(phrase, \text{"poor"})$$

Supervised classification [Pang et al., 2002]

- Classifiers
  - Naive Bayes
  - Maximum Entropy model
  - Support Vector Machine
- Features
  - N-grams (unigrams, bigrams)
  - POS tags
  - Negation
    - All words w_i that follow a negation term till a punctuation mark takes a form NOT_w_i
  - Positions
- Classification accuracy
  - Movie reviews: 82.9%
  - SVM

Sentiment Analysis in Tweets [Mohammad et al., 2013]

- Noisy and unstructured text from Twitter [Nakov et al., 2013]
  - Obama should be *impeached* on *TREASON* charges. Our Nuclear arsenal was TOP Secret. Till HE told our enemies what we had. *#Coward #Traitor*
  - My graduation speech: ”I’d like to *thanks* Google, Wikipedia and my computer! *:D* #iThingteens

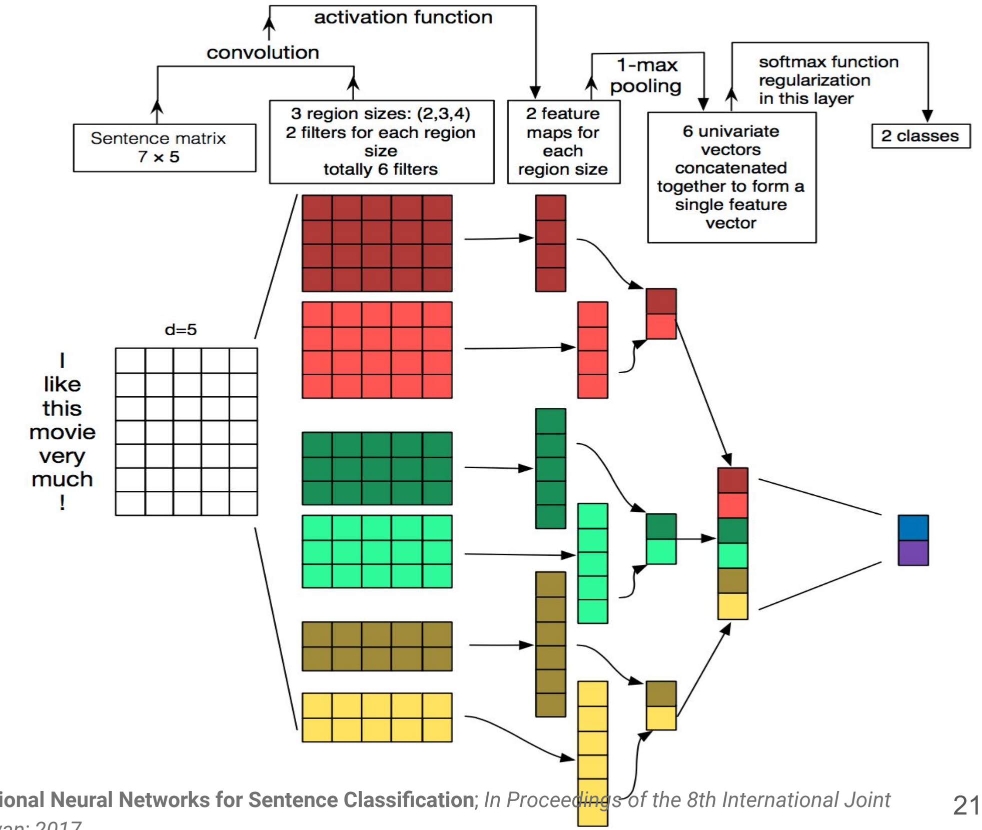
- Preprocessing
  - Urls and user ids → http://someurl and @someuser

- Features
  - Lexical and syntactical: Ngrams (word and character), All-caps, POS tags, negations
  - Twitter specifc: hashtags, punctuation, elongation, emoticons
  - Sentiment Lexicons: Bing Liu, MPQA, Sentiment140, Hashtag Sentiment Emoticons

- Classifier
  - SVM

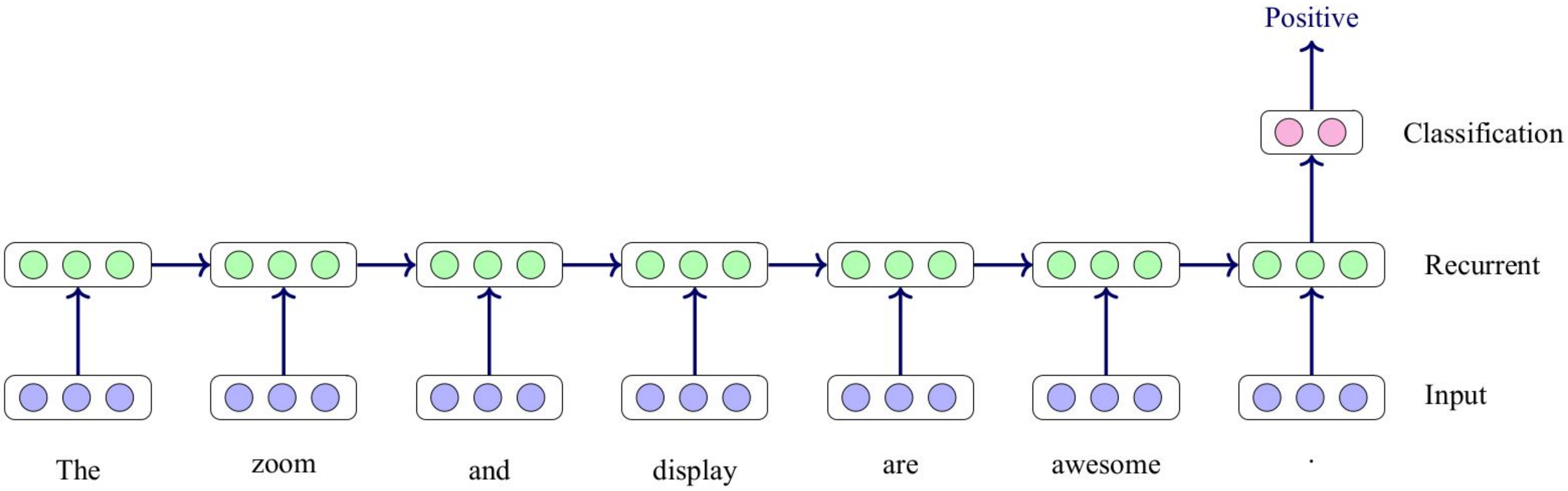
A Sensitivity Analysis of (and Practitioners' Guide to) Convolutional Neural Networks for Sentence Classification

1. Sentence matrix
   a. embeddings of words
2. Convolution filters
   a. Total 6 filters; Two each of size 2, 3 & 4.
   b. 6 feature maps for each filter
3. Pooling
   a. 1-max pooling
4. Concatenate the max-pooled vector
5. Classification
   a. Softmax



Zhang, Y., Wallace, B. ; A Sensitivity Analysis of (and Practitioners' Guide to) Convolutional Neural Networks for Sentence Classification; *In Proceedings of the 8th International Joint Conference on Natural Language Processing (IJCNLP-2017); pages 253-263; Taipie, Taiwan; 2017.*

Recurrent Model for Sentiment Analysis

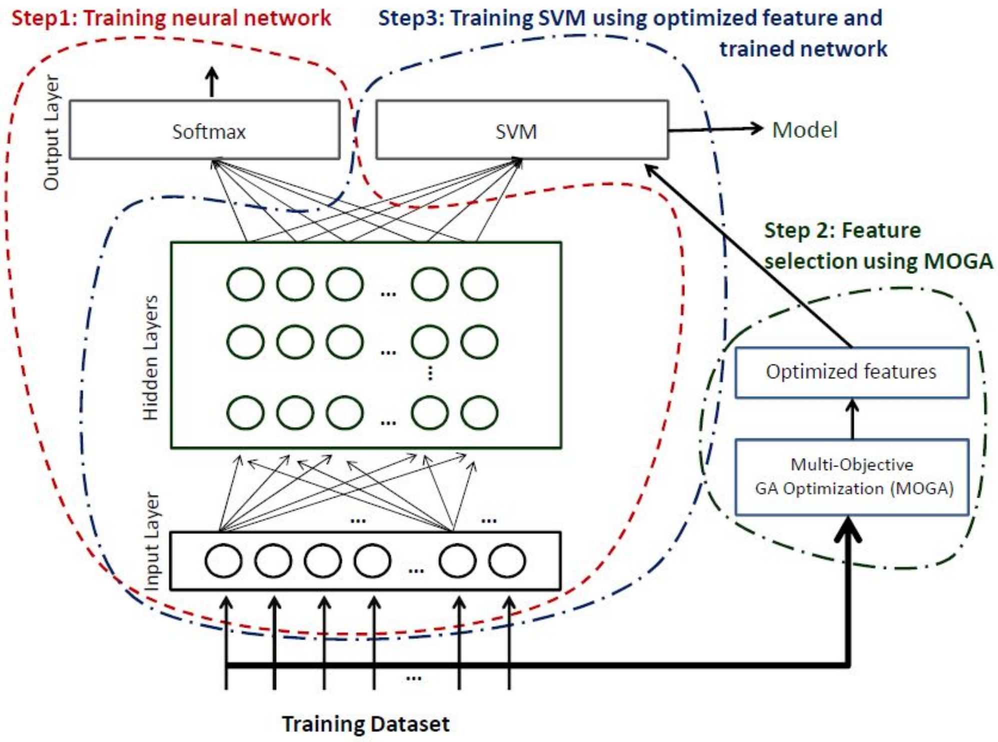


A Hybrid Deep Learning Architecture for Sentiment Analysis [Akhtar et al., 2016]

1. Training of a typical convolutional neural network (CNN)
    - Obtain weight matrix

2. A multi-objective GA based optimization technique (MOGA) for extracting the optimized set of features
    - Two objectives
        - *Accuracy* (maximize)
        - *Num of features* (minimize)
    - Optimized feature set

3. Training of SVM with non-linear kernel utilizing the network trained in first step and optimized features



Md Shad Akhtar, Ayush Kumar, Asif Ekbal and Pushpak Bhattacharyya; A Hybrid Deep Learning Architecture for Sentiment Analysis; In *proceedings of the 26th International Conference on Computational Linguistics (COLING 2016); 482–493; Osaka, Japan; 2016.*

A Multilayer Perceptron based Ensemble Technique for Fine-grained Financial Sentiment Analysis [Akhtar et al., 2017]

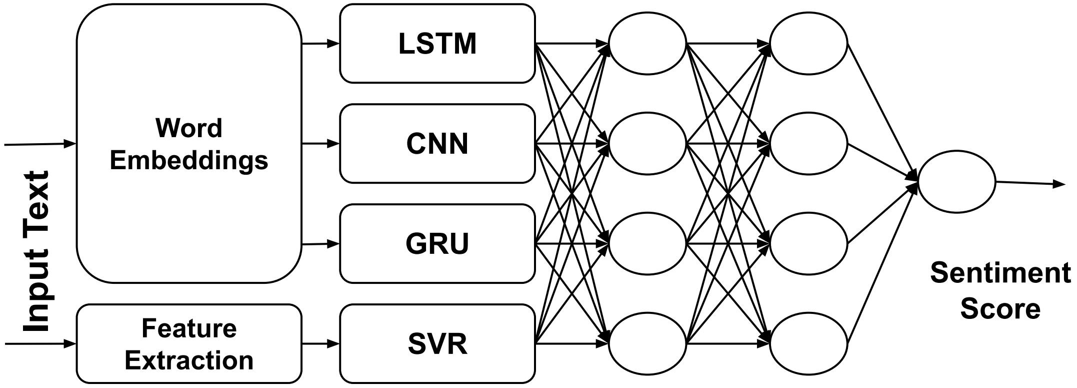
- Given a financial tweet, predict its sentiment score *w.r.t.* a company.

  best stock: $WTS +15%

    - Company/Cashtag: *WTS*
    - Sentiment: *Positive*
    - Intensity score: *0.857*

- Trained
  - Three DL methods:
    - LSTM, CNN & GRU
  - One feature driven
    - SVR
    - Tf-idf, lexicons

- Performance:
  - Numerically → Similar
  - Qualitative → Contrasting



Md Shad Akhtar, Abhishek Kumar, Deepanway Ghosal, Asif Ekbal and Pushpak Bhattacharyya; A Multilayer Perceptron based Ensemble Technique for Fine-grained Financial Sentiment Analysis; *In Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing (EMNLP 2017); 540 - 546; Copenhagen, Denmark; 2017.*

Aspect-based Sentiment Analysis (ABSA)

Sentiment Analysis: Granularity

- Based on the granularity of analysis, it can be categorized as belonging to one of the following four broad possible groups:
  - Document-level
  - Sentence-level
  - Phrase-level
  - Aspect-level

- **Aspect Based Sentiment analysis (ABSA):** Sentiment towards an aspect (or opinion-target or feature).

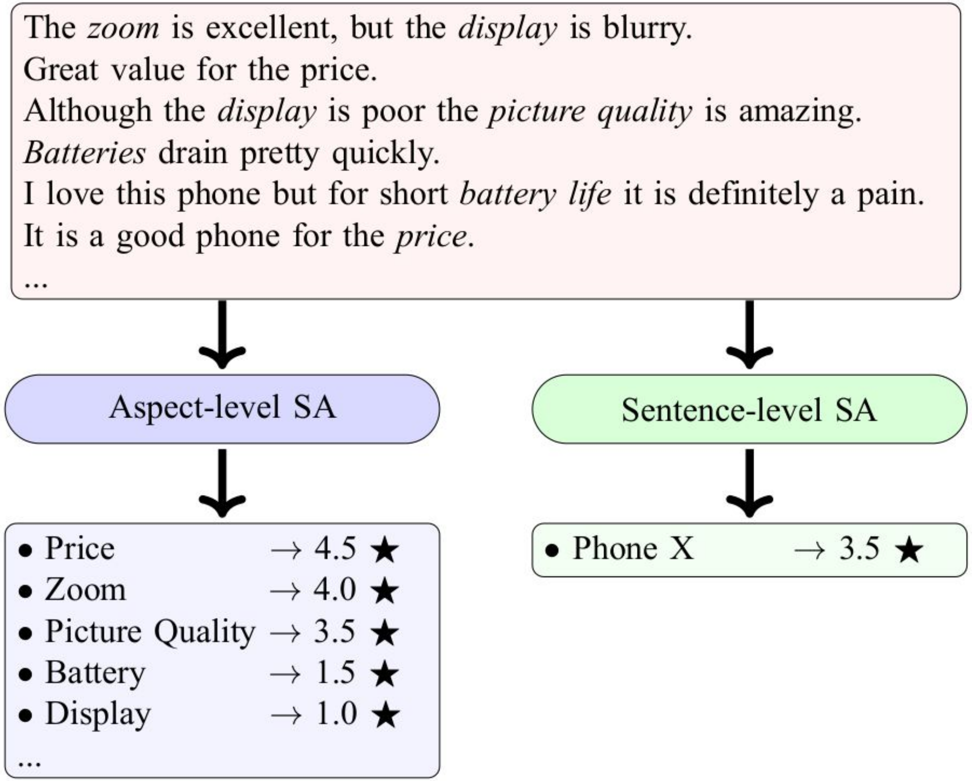
  Its battery is awesome but camera is very poor.

  इसकी बैटरी शानदार है, लेकिन कैमरा बहुत ही ख़राब है।

  (*Isakee baiTaree shaanadaara hai, lekin kaimaraa bahut hee kharaab hai..*)

  Positive about the ***battery*** but ***negative*** about the ***camera***

Coarse-grained vs Fine-grained Sentiment Analysis



SemEval shared tasks on ABSA [2014, 2015, 2016]

- **Aspect identification:** Type of aspects
- **Aspect-term (Opinion-target) extraction:** Finds the boundary of each aspect term present in the review.
- **Aspect sentiment classification:** Assigns a polarity value to each aspect term.

	Review Sentence	Aspect	Aspect Term	Sentiment
1:	The speed, the design.. it is lightyears ahead of any PC I have ever owned.	Hardware, Design	Speed, Design	Positive, Positive
2:	Tech support would not fix the problem unless I bought your plan for $150 plus.	Service	Tech support	Negative
3:	Certainly not the best sushi in New York, however, it is always fresh, and the place is very clean, sterile.	Food, Ambience	Sushi, Place	Conflict, Positive
4:	It was very expensive for what you get.	Price	No textual presence implies no aspect term	
5:	I enjoy having Apple products.	-No aspect or aspect term-		

Problem Formulation

- Aspect Term Extraction
    - Sequence labelling: Decide on each token whether its inside or outside of an aspect term
        - BIO encoding scheme

Although	the	display	is	poor	I	find	its	battery	life	amazing	.
O	O	B	O	O	O	O	O	B	I	O	O

- Aspect Sentiment Classification
    - Classification:
        - Context window size: ±2
            - *Although the display is poor* → 👎
            - *find its battery_life amazing.* → 👍
        - {Sentence, aspect} pairing
            - {*Although the display is poor I find its battery life amazing., display*} → 👎
            - {*Although the display is poor I find its battery life amazing., battery_life*} → 👍

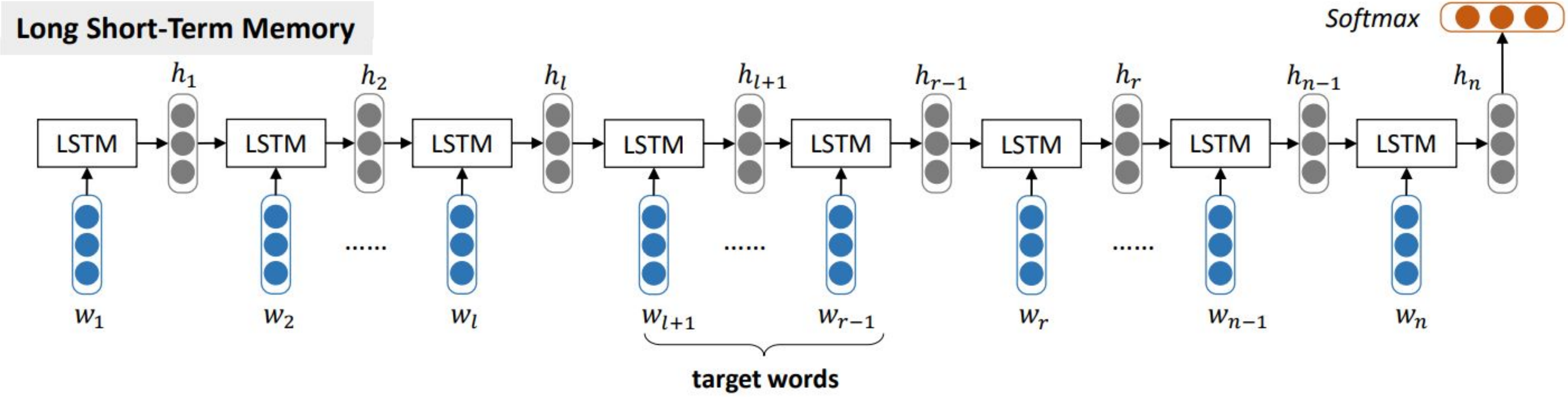
Effective LSTMs for Target-Dependent Sentiment Classification [Tang et al., 2016]

- Long Short-Term Memory (LSTM)
  - Models the semantic representation of a sentence without considering the target word being evaluated
- Target-Dependent Long Short-Term Memory (TD-LSTM)
  - Extend LSTM by considering the target word
- Target-Connection Long Short-Term Memory (TC-LSTM)
  - Semantic relatedness of target with its context words are incorporated

Duyu Tang, Bing Qin, Xiaocheng Feng, Ting Liu. 2016. Effective LSTMs for Target-Dependent Sentiment Classification. In Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers, pages 3298–3307, Osaka, Japan, December 11-17 2016.

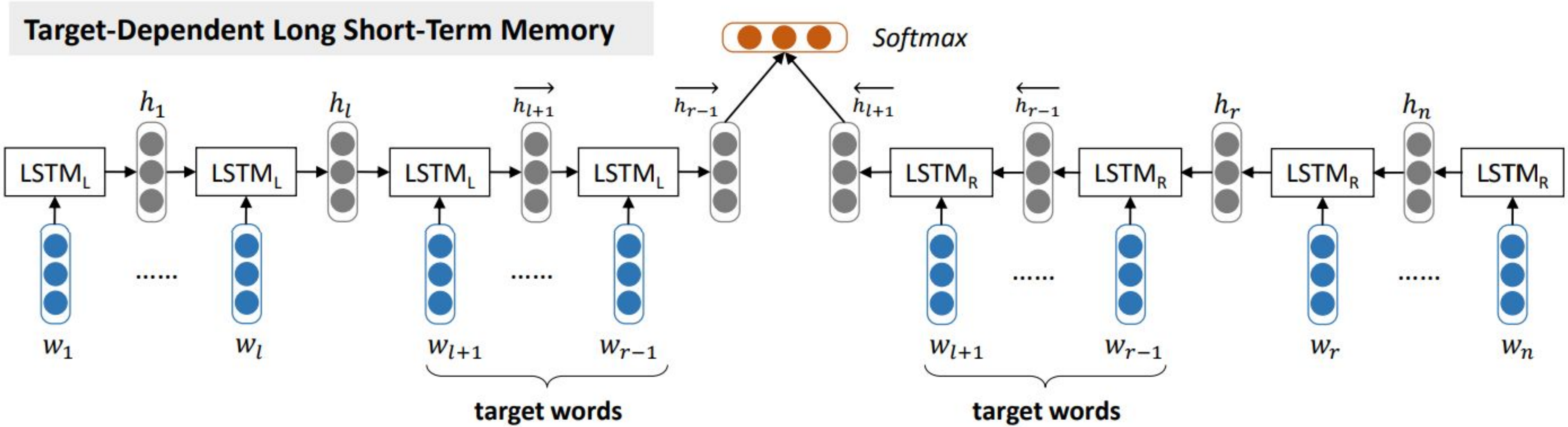
Simple LSTM [Tang et al., 2016]

- Models the semantic representation of a sentence without considering the target word being evaluated.
  - No discrimination between the following two instances
    - Its ***battery*** is awesome but camera is poor.
    - Its battery is awesome but ***camera*** is poor.



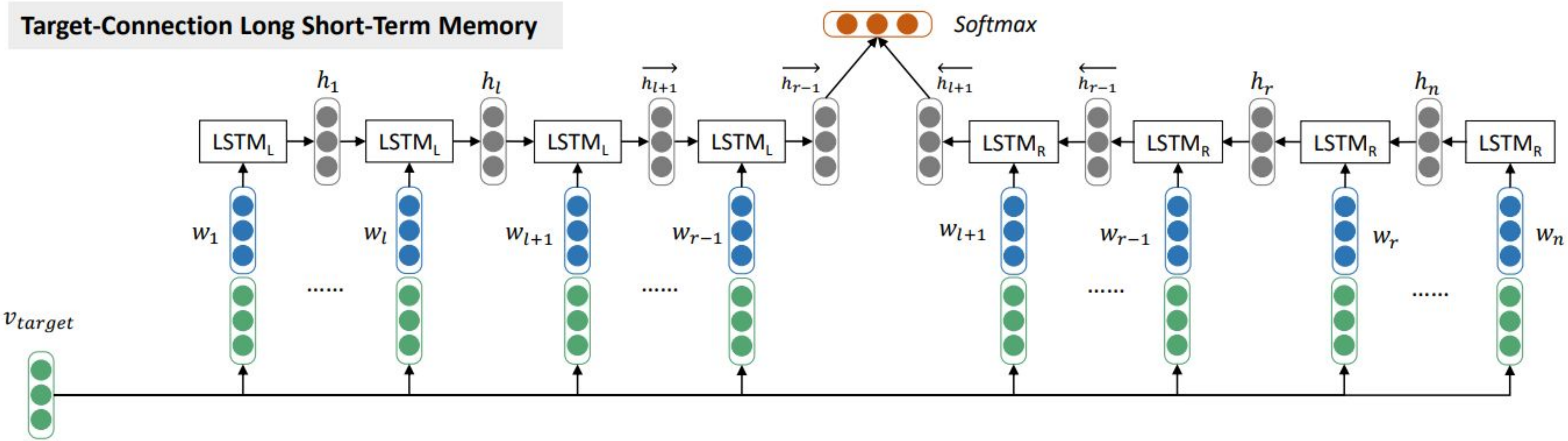
Target-Dependent Long Short-Term Memory (TD-LSTM) [Tang et al., 2016]

- Considers the target word
    - Its ***battery*** is awesome but camera is poor.
        - $LSTM_L$(Its ***battery***) + $LSTM_R$(***battery*** is awesome but camera is poor.)
    - Its battery is awesome but ***camera*** is poor.
        - $LSTM_L$(Its battery is awesome but ***camera***) + $LSTM_R$(***camera*** is poor.)



Target-Connection Long Short-Term Memory (TC-LSTM) [Tang et al., 2016]

- Relationship between the word and the target is incorporated
  - Its ***battery*** is awesome but camera is poor.
    - LSTM_{L1}(Its, ***battery***) → LSTM_{L2}(battery, ***battery***)
    - LSTM_{R7}(battery, ***battery***) ← LSTM_{R6}(is, ***battery***) ← LSTM_{R5}(awesome, ***battery***) ← LSTM_{R4}(but, ***battery***) ← LSTM_{R3}(camera, ***battery***) ← LSTM_{R2}(is, ***battery***) ← LSTM_{R1}(poor, ***battery***)



Available resources

- Datasets
  - English
    - IMDB movie review dataset http://www.cs.cornell.edu/people/pabo/movie-review-data/
    - Bing Liu review dataset https://www.cs.uic.edu/~liub/FBS/sentiment-analysis.html#datasets
    - Stanford Sentiment TreeBank (SST) https://nlp.stanford.edu/sentiment/treebank.html
    - Sentiment Analysis is Twitter https://www.cs.york.ac.uk/semeval-2013/task2.html
    - ABSA (SemEval-14, SemEval-15, SemEval-16) http://alt.qcri.org/semeval2014/task4/
  - Hindi
    - IIT Bombay (Hindi and Marathi) http://www.cfilt.iitb.ac.in/Sentiment_Analysis_Resources.html
    - IIT Patna (Hindi) http://www.iitp.ac.in/~ai-nlp-ml/resources.html
    - Sentiment Analysis in Indian Languages (Hindi, Bangla) http://amitavadas.com/SAIL/index.html